

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 23)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
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5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 1 . **1** Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

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0 1 . **2** Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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0 1 . **3** Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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10

0 2 . 1

Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

[4 marks]

0 2 . 2

Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

[3 marks]

0 2 . 3

Explain **three** ways in which coral reefs are important to humans.

[3 marks]

1

2

3

10

0 3 . 1 Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability

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Dissolved oxygen

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pH

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0 3 . 2 Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

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03 . 3

Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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10

04 . 1

Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar

☐

B Acoustic / satellite tag

☐

C Pitfall trap

☐

D Geolocator tag recovered by recapture

☐

04 . 2

Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 5

Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

Treatment	Mean dry mass / g	Standard deviation / g
Organic manure	14.2	3.9
Inorganic NPK fertiliser	16.1	4.1

0 5**1**

Explain what the standard deviation values tell you about the data.

[2 marks]

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0 5**2**

Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

[2 marks]

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0 5**3**

The test gave a probability $p = 0.30$. State what this means and what the student should conclude.

[2 marks]

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0 5**4**

Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

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0 5 . 5

Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

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0 6 . 1

Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

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0 6 . 2

Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

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06 . 3

Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

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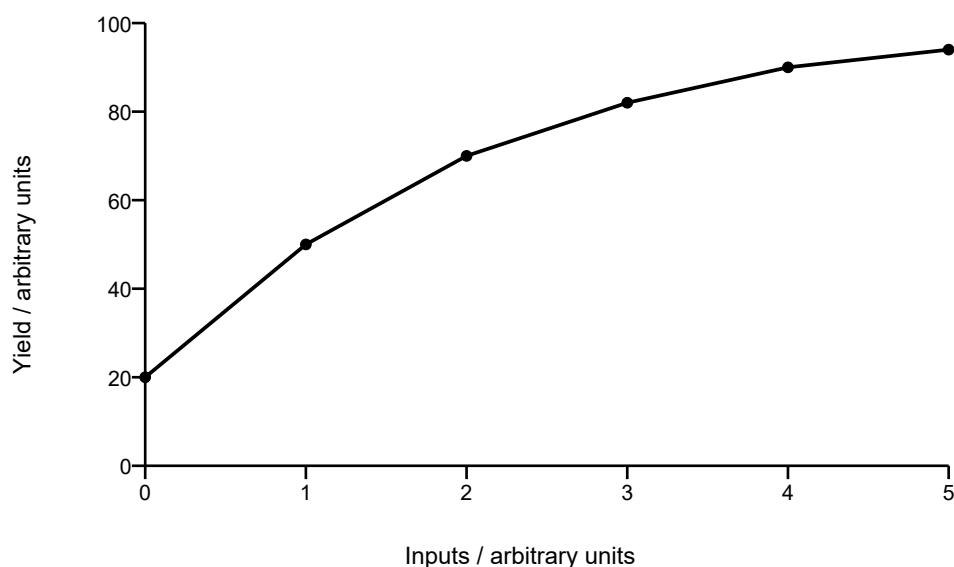
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10

07

Figure 1 shows the effect of increasing inputs (fertiliser, water and pesticides) on the yield of a crop.

Figure 1



07 . 1

Describe the relationship shown in Figure 1.

[2 marks]

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07 . 2

Calculate the extra yield produced by the second unit of input and by the fifth unit of input.

[2 marks]

Answer

units

07 . 3

Explain why, if enough land is available, spreading inputs evenly over a large area gives a greater total yield than concentrating them on a small area.

[2 marks]

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07 . 4

Suggest why inputs are nevertheless often used unevenly across the world.

[2 marks]

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07 . 5

Explain why intensive agriculture may be very productive but not very efficient.

[2 marks]

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08 . 1

Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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08 . 2

Describe **five** features of a suitable release site.

[5 marks]

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08 . 3

Explain the importance of post-release monitoring.

[2 marks]

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09 . 1

Explain why the ecosystem services provided by other species are often given a low conservation priority.

[2 marks]

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09 . 2

Explain how soil organisms provide services that humans depend on.

[3 marks]

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09 . 3

Explain why pollination by insects is more useful than wind pollination to plants growing in forests.

[2 marks]

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09 . 4

Explain how seed dispersal by animals increases the survival of a plant's offspring.

[3 marks]

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10 . 1

Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

Extra space

10 . 2 State what is meant by **industrial symbiosis**.

[1 mark]

10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

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Do not write outside the box

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Do not write outside the box

END OF QUESTIONS